

Nikolaos Giakoumidis

AI, Robotics and Automation | Engineering, Research and Technical Leadership

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PROFESSIONAL SUMMARY

Engineer, researcher and technical leader with 15+ years of experience spanning AI, robotics, electronics and automation. Led engineering projects, established and managed research laboratories, and brought people, resources and technologies together to turn ideas into working systems. Combines hands-on technical depth with responsibility for system architecture, integration, procurement, operations and multidisciplinary collaboration. Work with researchers across different fields has developed a broad scientific perspective and the ability to translate unfamiliar problems into practical engineering solutions. Currently leads commercialization and industry engagement at NYU Abu Dhabi's Center for Artificial Intelligence and Robotics, alongside PhD research in agentic robotics and physical AI.

EXPERIENCE

New York University Abu Dhabi (NYUAD) | 2012–Present

Commercial Lead – AI and Robotics | Center for Artificial Intelligence and Robotics (CAIR) | 2025–Present

- Lead commercialization and industry engagement for CAIR, working with faculty, researchers and external partners to identify practical needs and develop applications for AI, robotics and autonomous systems.
- Authored 15+ project proposals in the first year, translating industrial needs and research capabilities into proposed work across inspection, logistics, infrastructure, digital twins and autonomous systems.
- Completed five NDAs with external stakeholders, enabling further technical exchange and collaboration discussions.
- Connect scientific teams with industry and government partners, bringing engineering judgment to discussions of technical feasibility, implementation needs and potential applications.
- Designed and built a model-agnostic robot agent harness, MCP skill servers and orchestration for natural-language industrial inspection, connecting AI reasoning with robot sensing and actions within a deterministic safety boundary.

- Represent NYUAD and CAIR through technical demonstrations, executive discussions and exhibitions including ADIPEC, Global Rail, Make it in the Emirates and DriftX.
- Engage organizations across transport, energy, logistics, technology and investment, including AD Ports Group, Etihad Rail, GCAA, Kent, Mubadala, MGX, Analog, CycloTech and Mimik.

Research Instrumentation Specialist – Robotics & Automation | Core Technology Platforms (CTP) | 2017–2025

- Established, expanded and operated laboratories across Robotics, Photonics, High-Throughput Screening, Electronics and Advanced Manufacturing, building shared engineering capabilities for multidisciplinary research.
- Led end-to-end technical decision-making across multiple research laboratories, covering facility design, construction supervision, system architecture, integration and commissioning, equipment strategy, vendor selection and procurement, workflow design, safety procedures, and repair-versus-replacement decisions.
- Supported 100+ users, including faculty, researchers, students and technical staff, translating experimental requirements into equipment configurations, integrated systems and working research methods.
- Stewarded US\$9M+ in research assets and influenced approximately US\$800K in annual purchasing, with peak periods approaching US\$3M, through technical evaluation, vendor selection and procurement planning.
- Diagnosed and restored complex scientific equipment, including systems without vendor support, to maintain research continuity and avoid unnecessary service or replacement expenditure.
- Co-developed and commissioned the Photonics laboratory and managed its operation and expansion. Engineered HTS automation through Thermo Fisher Momentum workflows, FANUC configuration, instrument synchronization and fault recovery.
- Organized approximately four student workshops per semester and co-supervised around two capstone projects annually, mentoring users in engineering methods, system integration and experimental work.
- Operated within demanding safety and regulatory environments involving drones, lasers, and RF systems, while maintaining compliance with university and UAE approval requirements.
- Supported multidisciplinary research output across NYUAD, with technical contributions and infrastructure support recognized in publications and acknowledgements across multiple research domains.

Electronics Engineer, Laboratories | 2013–2017

- Established the Electronics Workshop and co-established the Advanced Manufacturing Workshop, expanding in-house capabilities in prototyping, scientific instrumentation and engineering support.

- Played a key role in planning and relocating research laboratories from the Center for Science and Engineering to the Saadiyat campus.
- Designed and integrated electronics, control hardware and automation for research teams with different scientific specialisms, supporting experimental development from problem definition through testing.
- Supported the operation and technical development of the High Throughput Screening Platform, integrating robotic automation with scientific instruments and laboratory workflows.
- Delivered in-house troubleshooting and repair of scientific equipment, supporting reliable laboratory operation and informed repair-versus-replacement decisions.
- Presented laboratory capabilities and working systems to industry and government delegations, translating technical research into accessible demonstrations.

Research Assistant | Interactive Robots and Media Laboratory (IRML) | 2012–2013

- Contributed to the establishment and early operation of NYUAD's first robotics laboratory.
- Developed and integrated robotic systems for research, including work linking aerial mapping with ground-robot path planning, and contributed to related publications.
- Served as a teaching assistant for engineering courses and supported international robotics workshops.

Hellenic Armed Forces, Dept. of Advanced Defense Systems | 2011–2012

R&D Engineer | Mandatory national service in Greece

- Conducted R&D on small unmanned aerial systems, including fixed-wing and multirotor platforms.

Experimental Primary School of Athens | 2010–2011

Teacher of Robotics

- Delivered robotics instruction to school students using LEGO Mindstorms, supporting early STEM engagement.

United Arab Emirates University, Al Ain | 2009–2010

Research Intern | Interactive Robots and Media Lab

- Supported robotics research activities and project execution in the Interactive Robots and Media Lab.

G&D Goumas SA, Athens | 2003–2009

Manager of Industrial and Automotive Equipment

- Managed technical support and training activities for industrial and automotive diagnostic equipment, including Launch diagnostic tools and troubleshooting systems.

EDUCATION

University of the Aegean, Greece

PhD Candidate | Started 2023 | Expected completion: January 2027

- Research on autonomous cooperative robotic agents, embodied AI and deep reinforcement learning. Current work develops agent harnesses connecting AI models with sensing, control, reasoning, planning and coordination in physical systems.

University of West Attica, Greece

BSc, Automation Engineering | 2012

SELECTED CERTIFICATIONS

- Dubai Civil Aviation Authority | Professional RPAS Certificate, Multirotor & Fixed Wing, 2023
- NVIDIA Deep Learning Institute | Deep Learning for Robotics, 2019
- KUKA College | KUKA LBR iiwa Commissioning and Programming, 2019
- Udacity | Deep Reinforcement Learning Nanodegree, 2019

AWARDS

- **Team First Prize**, RTA Dubai World Challenge for Self-Driving Transport | Delivery Drone, 2021
- **First Prize**, Design EmbedIT Competition, Shanghai, 2012
- **First Prize**, World Robot Olympiad, Greek national round, 2009
- **First Prize**, Digital Week Robotics Competition, Athens, 2008

CORE COMPETENCIES

AI, Robotics & Autonomous Systems

Agentic AI, embodied and physical AI, multi-agent robotic systems, multimodal perception, human–robot interaction, autonomous inspection, control, localization, mapping, navigation,

path planning, teleoperation; ground, aerial and underwater robotics; electronics, mechatronic integration and scientific instrumentation.

Programming, AI & Deployment

Python, C++, C, MATLAB and LabVIEW; ROS 2, MCP, LLM/VLM workflows, local inference with Ollama, PyTorch, Docker, FastAPI and n8n; supporting experience with TensorFlow, CUDA, TensorRT, Hugging Face, LangChain and retrieval-augmented generation.

Platforms, Simulation & Control

Isaac Sim, Isaac Lab, Gazebo and MATLAB Simulink; PX4, ArduPilot and DJI; Boston Dynamics Spot, Clearpath Husky, Unitree H1/G1, KUKA, FANUC, Kinova and ST Robotics. Sensing and instrumentation include LiDAR, radar, RGB/thermal and multispectral cameras, motion capture, GNSS, IMUs, 3D scanners, optical and RF systems.

Engineering Leadership and Research Operations

System architecture, technical assessment, requirements definition, laboratory establishment, integration and commissioning, equipment strategy, procurement, vendor coordination, resource planning, safety procedures, fault diagnosis, mentoring and user training.

Industry Engagement and Research Translation

Industry needs discovery, multidisciplinary research collaboration, proposal development, technical demonstrations, stakeholder communication and assessment of potential applications across government and industry.

SELECTED PUBLICATIONS

- Giakoumidis, N., Anagnostopoulos, C.-N. "ARM4CH: A Methodology for Autonomous Reality Modelling for Cultural Heritage." *Sensors*, 2024.
- Prieto, S. A., Giakoumidis, N., García de Soto, B. "Multiagent Robotic Systems and Exploration Algorithms: Applications for Data Collection in Construction Sites." *Journal of Field Robotics*, 41(4), 1187–1203, 2024.
- Eid, M., Giakoumidis, N., El Saddik, A. "A Novel Eye-Gaze-Controlled Wheelchair System for Navigating Unknown Environments." *IEEE Access*, 2016.
- Tsoukalas, A., Xing, D., Evangeliou, N., Giakoumidis, N., Tzes, A. "Deep Learning Assisted Visual Tracking of Evader-UAV." *ICUAS*, 2021.
- Xing, D., Tsoukalas, A., Evangeliou, N., Giakoumidis, N., Tzes, A. "Siamese Adaptive Transformer Network for Real-Time Aerial Tracking." *ICUAS*, 2022.
- Giakoumidis, N., Bak, J. U., Gómez, J. V., Llenga, A., Mavridis, N. "Pilot-Scale Development of a UAV-UGV Hybrid with Air-Based UGV Path Planning." In: 2012 10th International Conference on Frontiers of Information Technology, pp. 204–208. IEEE, 2012.

Full publication list available via Google Scholar:

<https://scholar.google.com/citations?user=HmOOogwAAAAJ&hl=en>